

BRIDGE COURSE

GRADE – 8

Biology

Grade 7

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1. LIFE AS AN ORGANIZED BIOLOGICAL SYSTEM

1.1 Defining Life Scientifically

Life is defined by a set of essential characteristics that distinguish living organisms from non-living matter. All living beings exhibit cellular organization, meaning their bodies are composed of one or more cells.

Inside these cells, continuous biochemical reactions occur, collectively known as metabolism. These reactions include processes such as digestion, respiration, synthesis of new molecules, and waste removal.

Growth in living organisms is not merely an increase in size; it involves cell division and an increase in internal complexity. Reproduction allows organisms to produce offspring and ensure continuity of their species. Another key feature of life is response to stimuli organisms react to changes in their environment, such as light, temperature, sound, or chemicals.

Adaptation and evolution over long periods further define living systems. Thus, life is maintained through organization, metabolism, growth, reproduction, responsiveness, and adaptation.

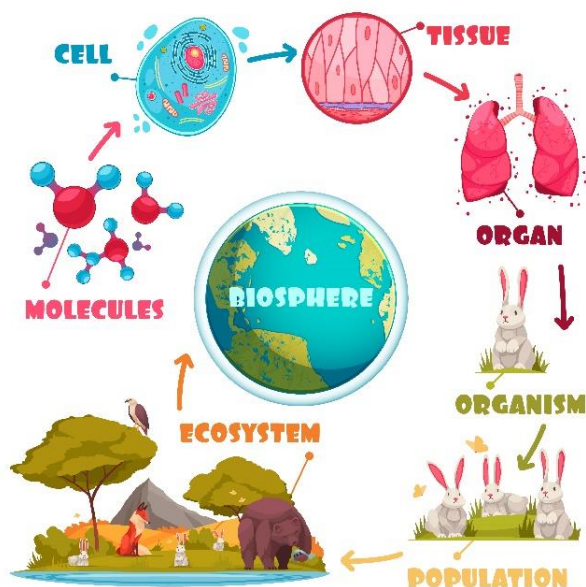


1.2 Cellular Basis of Life

The cell is the structural and functional unit of life. All organisms are composed of cells. In unicellular organisms, a single cell performs all life functions such as nutrition, respiration, excretion, and reproduction.

In multicellular organisms, cells are specialized to perform specific functions. Similar cells form tissues, tissues form organs, and organs form organ systems. This hierarchical organization increases efficiency and allows complex life forms to function properly.

Cells contain cytoplasm, genetic material (DNA), and specialized structures called organelles that perform distinct functions. The coordination among billions of cells enables organisms to maintain stability and survive in changing environments.



Examples

(i) **“Life cannot be defined by movement alone.” Justify this statement**

Solution: Life cannot be defined only by visible movement because many living organisms do not show obvious movement. For example, plants remain fixed in one place, yet they are living. Even microscopic organisms may not be visible to the naked eye, but they perform all life processes internally. The true definition of life lies in cellular organization and metabolism. Inside every living cell, continuous biochemical reactions occur, such as respiration, synthesis of molecules, and waste removal. Growth through cell division, reproduction for species continuity, and response to stimuli further define life. Therefore, life is characterized by internal biological activity rather than external motion alone.

(ii) **Describe the hierarchical organization in multicellular organisms.**

Solution: In multicellular organisms, cells are organized in a hierarchical manner. Similar cells performing the same function group together to form tissues. Different tissues combine to form organs. Several organs work together to form organ systems, such as the digestive system or circulatory system. This hierarchical organization increases efficiency and allows complex organisms to perform various life processes smoothly. Coordination among different organ systems maintains stability and survival of the organism.

(iii) **Assertion (A): Life is maintained through organization, metabolism, growth, reproduction, responsiveness, and adaptation.**

Reason (R): These characteristics distinguish living organisms from non-living matter.

(A) Both A and R are true, R is correct explanation of A

(B) Both A and R are true, R is not correct explanation of A

(C) A is true, R is false

(D) A is false, R is true

Solution: (A)

Both statements are true because the listed features are essential characteristics of life. These traits distinguish living organisms from non-living matter, so the reason correctly explains the assertion.

(iv) **In a unicellular organism, a single cell performs:**

(A) Only reproduction

(B) All life functions including nutrition, respiration, excretion, and reproduction

(C) Only metabolism

(D) Only response to stimuli

Solution: (B)

A unicellular organism has only one cell, so it must perform all vital functions like nutrition, respiration, excretion, and reproduction. Hence, the single cell carries out all life processes.

(v) **Which of the following correctly explains the importance of cellular specialization?**

(A) It allows efficiency and proper functioning of complex organisms

(B) It prevents cells from performing life functions

(C) It occurs only in unicellular organisms

(D) It eliminates the need for tissues

Solution: (A)

Cellular specialization allows different cells to perform specific functions efficiently in multicellular organisms. This division of labour ensures proper functioning of complex bodies.



Practice Questions - 1

- (i) Which of the following is common to both unicellular and multicellular organisms?

(A) Presence of tissues	(B) Presence of organs
(C) Cellular organization	(D) Division of labour
- (ii) A crystal increases in size when placed in a saturated solution. Why is this not considered biological growth?

(A) It lacks visible movement	(B) It does not involve cell division
(C) It cannot reproduce	(D) It does not require water
- (iii) The primary role of DNA in a cell is to:

(A) Store food	(B) Produce energy directly
(C) Control cellular activities	(D) Transport oxygen
- (iv) Assertion (A): Movement is not a reliable criterion for defining life.
Reason (R): Some non-living objects such as vehicles can move.

(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.
- (v) Which statement is scientifically correct?

(A) All living organisms show locomotion.
(B) All living organisms are made up of at least one cell.
(C) All cells have identical functions.
(D) Growth alone defines life.
- (vi) Which situation best demonstrates that life depends on continuous chemical activity?

(A) A rock remaining unchanged for years	(B) A plant synthesizing food
(C) A machine producing sound	(D) A cloud changing shape
- (vii) Why is reproduction essential for species survival but not necessarily for individual survival?
- (viii) Why is movement not considered a reliable criterion for defining life?
- (ix) Distinguish between unicellular and multicellular organisms based on organization and function.
- (x) "Life is a balance between structure and continuous chemical activity." Elaborate with reasoning.

2. ENERGY FLOW: FOUNDATION OF BIOLOGICAL EXISTENCE

2.1 Sun as the Primary Energy Source

The Sun is the ultimate source of energy for almost all life on Earth. Green plants capture solar energy using chlorophyll in their leaves. Through the process of photosynthesis, they convert carbon dioxide and water into glucose and oxygen.

Glucose stores chemical energy, which becomes available to other organisms when they consume plants.

Oxygen released during photosynthesis supports respiration in living organisms. Without solar energy and photosynthesis, food chains would collapse and atmospheric oxygen levels would decline.

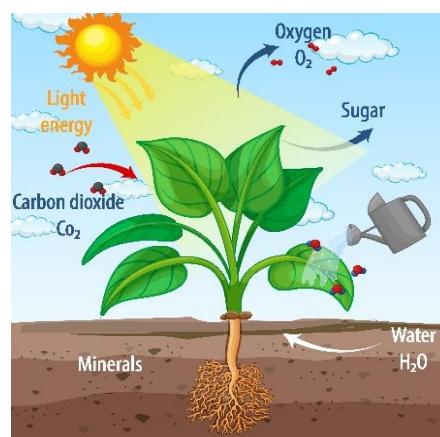


2.2 Photosynthesis as Energy Conversion and Carbon Fixation

Photosynthesis is both an energy transformation and a carbon fixation process. During this process, light energy is converted into chemical energy stored in glucose molecules.

Carbon dioxide from the atmosphere is incorporated into organic compounds, forming the basis of all food substances.

This process maintains the balance of oxygen and carbon dioxide in the atmosphere. It also plays an important role in regulating global climate by reducing excess carbon dioxide.

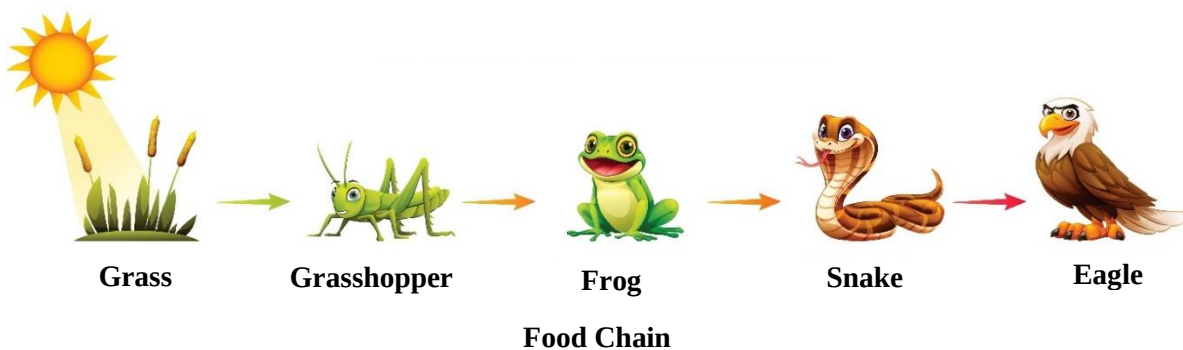


2.3 Energy Transfer Through Food Chains

Energy moves through ecosystems in a unidirectional flow. Producers (green plants) form the first trophic level.

Herbivores that eat plants form the second trophic level, while carnivores and top predators occupy higher trophic levels. At each level, only a small portion of energy is transferred to the next level; most of it is lost as heat during respiration.

This loss of energy explains why food chains are short and why the number of organisms decreases at higher trophic levels. Energy pyramids visually represent this decrease in available energy.



Examples

(i) Explain how solar energy is transformed into a usable form of energy for living organisms.

Solution: Green plants capture solar energy through chlorophyll during photosynthesis. This light energy is converted into chemical energy and stored in glucose molecules. When organisms consume plants, this stored chemical energy becomes available to them through respiration. Thus, solar energy is transformed into a usable form for all living beings.

(ii) Explain why energy cannot be recycled in ecosystems, whereas nutrients can.

Solution: Energy flows in one direction from the Sun to producers and then to consumers and is ultimately lost as heat. Once lost, it cannot be reused by organisms.

Nutrients like carbon and nitrogen, however, circulate through biogeochemical cycles and are reused repeatedly by living organisms. Therefore, energy flow is non-cyclic, while nutrient flow is cyclic.

(iii) Assertion (A): Energy flow in ecosystems is cyclic.

Reason (R): Nutrients are recycled through ecosystems.

(A) Both A and R are true, and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, but R is false.

(D) A is false, but R is true.

Solution: (D)

Energy flow in ecosystems is unidirectional, not cyclic, so the assertion is false. However, nutrients are recycled through biogeochemical cycles, making the reason true.

(iv) Arrange the following in correct order of energy flow:

1. Carnivores

2. Sun

3. Producers

4. Herbivores

(A) 2 → 3 → 4 → 1

(B) 3 → 2 → 4 → 1

(C) 2 → 4 → 3 → 1

(D) 3 → 4 → 1 → 2

Solution: (A)

Energy flows from the Sun → Producers → Herbivores → Carnivores. This sequence correctly represents the transfer of energy across trophic levels in a food chain.

(v) Cause: Increased carbon dioxide levels

Effect: Increased photosynthetic activity (initially)

The effect occurs because:

(A) Carbon dioxide is a product of respiration

(B) Carbon dioxide is a raw material for photosynthesis

(C) Oxygen is reduced

(D) Energy flow becomes cyclic

Solution: (B)

Carbon dioxide is a raw material used by plants during photosynthesis. Therefore, an increase in CO₂ initially increases photosynthetic activity.



Practice Questions - 2

- (i) Which statement is scientifically correct?
 - (A) Energy can be reused indefinitely in ecosystems
 - (B) Photosynthesis and respiration are unrelated processes
 - (C) Producers determine the energy availability in ecosystems
 - (D) Energy increases as it moves up trophic levels
- (ii) If the producer level in an ecosystem declines sharply, which would be the immediate effect?
 - (A) Increase in carnivore population
 - (B) Increase in atmospheric oxygen
 - (C) Decrease in herbivore population
 - (D) Decrease in decomposition rate
- (iii) Which statement correctly describes energy flow in an ecosystem?
 - (A) Energy is recycled through food chains
 - (B) Energy flows in a cyclic manner
 - (C) Energy flows in one direction and decreases at each level
 - (D) Energy increases at higher trophic levels
- (iv) Which scenario would most likely cause collapse of food chains?
 - (A) Increase in herbivore population
 - (B) Increase in solar radiation
 - (C) Sudden disappearance of green plants
 - (D) Increase in rainfall
- (v) Assertion (A): Energy pyramids are always upright.
Reason (R): Energy decreases at each trophic level due to heat loss during respiration.
 - (A) Both A and R are true, and R is the correct explanation of A.
 - (B) Both A and R are true, but R is not the correct explanation of A.
 - (C) A is true, but R is false.
 - (D) A is false, but R is true.
- (vi) A decrease in producer population would most immediately affect:
 - (A) Atmospheric nitrogen
 - (B) Herbivores
 - (C) Top predators only
 - (D) Decomposers only
- (vii) If atmospheric carbon dioxide levels increased significantly, how might photosynthesis influence global climate?
- (viii) Evaluate the importance of photosynthesis in sustaining ecological balance and regulating climate.
- (ix) Describe how carbon fixed during photosynthesis moves through an ecosystem.
- (x) Describe the process of energy flow in an ecosystem. Why is energy transfer between trophic levels inefficient?

3. RESPIRATION AND ENERGY LIBERATION

3.1 Difference Between Breathing and Respiration

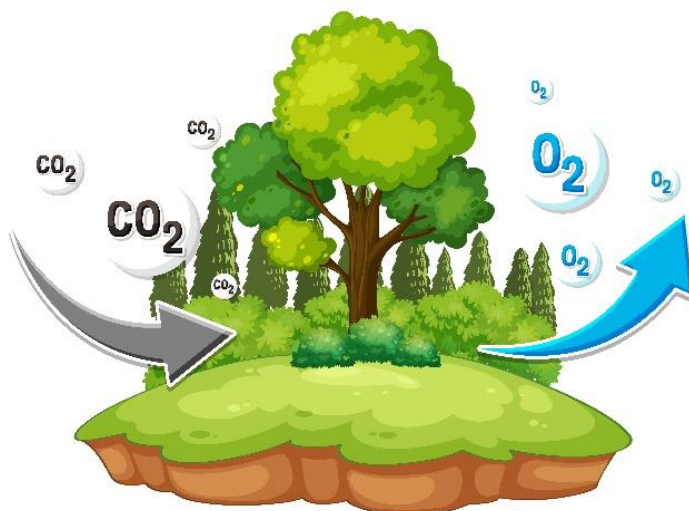
Breathing is a physical process involving the intake of oxygen and release of carbon dioxide through the respiratory organs. Respiration is a chemical process occurring inside cells where glucose is broken down to release energy. This energy is stored in ATP (adenosine triphosphate) molecules.

Aerobic respiration requires oxygen and produces carbon dioxide, water, and a large amount of energy. Anaerobic respiration occurs in the absence of oxygen and releases less energy. Understanding this distinction is important because breathing only supplies oxygen, while respiration actually releases usable energy for cellular activities.

3.2 Role of Oxygen in Supporting Complex Life

Oxygen allows complete breakdown of glucose during aerobic respiration, producing maximum energy. This high energy yield supports complex activities such as movement, thinking, growth, and repair.

In the absence of oxygen, organisms rely on less efficient anaerobic processes, which cannot support highly complex structures for long periods.



Examples

(i) **Explain why complex organisms cannot depend only on anaerobic respiration.**

Solution: Anaerobic respiration releases less energy because glucose is not completely broken down. Complex organisms require large amounts of energy for activities like thinking and movement. Therefore, anaerobic respiration alone cannot meet their high energy demands.

(ii) **Two cell samples are observed:**

Sample A shows complete breakdown of glucose and high ATP production.

Sample B shows partial breakdown and low ATP production.

Identify the type of respiration in both samples and justify.

Solution: Sample A undergoes aerobic respiration because glucose is completely broken down, producing a large amount of ATP.

Sample B undergoes anaerobic respiration because glucose is partially broken down, producing less ATP.

Aerobic respiration requires oxygen and produces maximum energy, whereas anaerobic respiration occurs without oxygen and produces less energy.

(iii) **Which statements justify that breathing alone cannot sustain life?**

1. Breathing does not produce ATP.

2. Energy release occurs inside cells.

3. Oxygen is useless without respiration.

4. Respiration requires glucose breakdown.

(A) 1, 2 and 4 only

(B) 1 and 3 only

(C) 2, 3 and 4 only

(D) 1, 2, 3 and 4

Solution: (D)

Breathing alone cannot sustain life because it only provides oxygen, which is necessary for cellular respiration but does not produce ATP directly. ATP, the energy required for cell functions, is generated through respiration, which involves the breakdown of glucose inside cells. Without respiration, oxygen alone cannot fuel the body's energy needs.

(iv) **A cell is supplied with glucose but deprived of oxygen. What will most likely happen?**

(A) Maximum ATP production

(B) No respiration

(C) Increased carbon dioxide production

(D) Reduced ATP formation

Solution: (D)

Without oxygen, cells shift to anaerobic respiration, which produces much less ATP than aerobic respiration. Hence, ATP formation is reduced.

(v) **Which situation best represents respiration but not breathing?**

(A) Air entering lungs

(B) Oxygen moving through nose

(C) Carbon dioxide leaving body

(D) Glucose breaking down inside cells

Solution: (D)

Respiration refers to the biochemical process in which glucose is broken down inside cells to produce energy (ATP), while breathing is the physical process of inhaling oxygen and exhaling carbon dioxide.



Practice Questions - 3

- (i) The efficiency of energy production in cells primarily depends on:

(A) Speed of inhalation	(B) Availability of oxygen
(C) Size of lungs	(D) Amount of carbon dioxide
- (ii) Statement I: Complex organisms require oxygen continuously.
Statement II: Oxygen enables complete breakdown of glucose.

(A) Both statements are true and II explains I.
(B) Both statements are true but II does not explain I.
(C) I is true but II is false.
(D) I is false but II is true.
- (iii) The fundamental reason respiration is classified as a chemical process is because it:

(A) Involves inhalation	(B) Occurs in respiratory organs
(C) Breaks down glucose to release energy	(D) Requires lungs
- (iv) Energy released during respiration is stored in the form of:

(A) Oxygen	(B) Carbon dioxide
(C) ATP	(D) Water
- (v) Statement I: Breathing alone can maintain life.
Statement II: Energy release occurs during respiration inside cells.

(A) Both statements true	(B) I true, II false
(C) I false, II true	(D) Both false
- (vi) Which of the following correctly describes respiration?

(A) Exchange of gases in lungs	(B) Physical movement of air
(C) Breakdown of glucose to release energy	(D) Intake of oxygen only
- (vii) A student runs fast during a race. His breathing rate increases. His muscles require more energy for movement. Why does breathing rate increase during running?
- (viii) How does breathing help in the process of respiration?
- (ix) If oxygen supply becomes insufficient, what type of respiration will occur and why is it less efficient?
- (x) A researcher observes two organisms:
Organism A survives only in the presence of oxygen and performs active movements.
Organism B survives even without oxygen but shows limited activity.
Explain the difference in their energy production.

4. MATTER CYCLING IN NATURE

4.1 Water Cycle

The water cycle describes the continuous movement of water between Earth's surface and atmosphere. Water evaporates from oceans, lakes, and rivers due to solar heat. It condenses into clouds and returns as precipitation such as rain or snow. Water infiltrates soil or flows into water bodies again. This cycle ensures constant availability of fresh water. Water is essential for digestion, transportation of nutrients, temperature regulation, and photosynthesis.



4.2 Carbon Cycle

The carbon cycle maintains the balance of carbon in nature. Plants absorb carbon dioxide during photosynthesis and convert it into organic compounds. Animals obtain carbon by consuming plants or other animals.

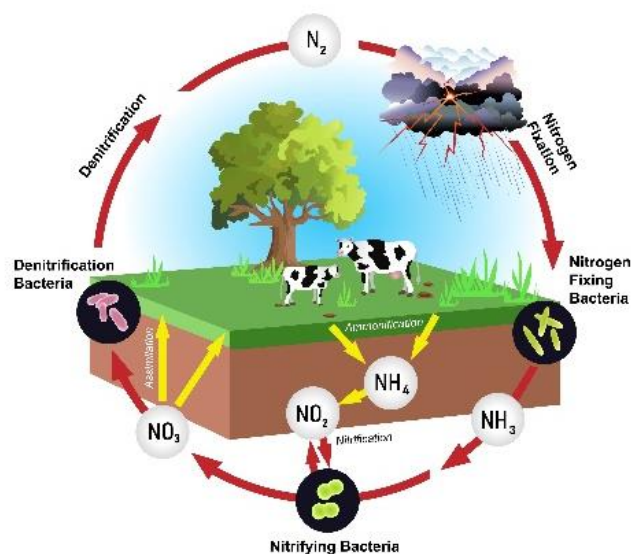
Respiration releases carbon dioxide back into the atmosphere. Decomposers break down dead organisms, returning carbon to the environment. Burning fossil fuels adds additional carbon dioxide, influencing climate patterns.

4.3 Nitrogen Cycle

Nitrogen is abundant in the atmosphere but cannot be directly used by most organisms. Nitrogen-fixing bacteria convert atmospheric nitrogen into usable forms like nitrates and ammonia.

Plants absorb these compounds and incorporate them into proteins. Animals obtain nitrogen by feeding on plants or other animals.

Decomposition returns nitrogen to the soil, and certain bacteria convert it back to atmospheric nitrogen. This continuous cycle maintains soil fertility and supports life.



Examples

(i) **How do plants act as a link between the atmosphere and living organisms in the carbon cycle?**

Solution:

- Plants absorb carbon dioxide from the atmosphere during photosynthesis.
- They convert it into organic compounds.
- Animals obtain carbon by consuming plants, thus transferring carbon through the food chain.

(ii) **Explain how the three cycles are interdependent in supporting life.**

Solution:

- The water cycle ensures availability of water required for digestion, nutrient transport, temperature regulation, and photosynthesis.
- The carbon cycle supplies carbon dioxide for photosynthesis and returns carbon through respiration and decomposition.
- The nitrogen cycle provides nitrogen compounds for protein formation.
- Together, these cycles maintain balance in nature and support life processes.

(iii) **Assertion (A): Burning fossil fuels increases atmospheric carbon dioxide.**

Reason (R): Fossil fuels contain stored carbon compounds.

(A) Both A and R are true, and R explains A

(B) Both A and R are true, but R does not explain A

(C) A is true, R is false

(D) A is false, R is true

Solution: (A)

Burning fossil fuels releases carbon dioxide because these fuels are made of stored carbon compounds formed from ancient organisms. Hence, both statements are true and the reason correctly explains the assertion

(iv) **Atmospheric nitrogen → _____ → Plants → Animals**

(A) Respiration

(B) Nitrogen-fixing bacteria

(C) Evaporation

(D) Decomposition

Solution: (B)

Atmospheric nitrogen cannot be used directly by plants. Nitrogen-fixing bacteria convert it into usable forms (like nitrates), which then enter plants and subsequently animals.

(v) **Identify the incorrect statement:**

(A) Photosynthesis removes carbon dioxide from air.

(B) Nitrogen-fixing bacteria increase soil fertility.

(C) Precipitation increases atmospheric water vapor.

(D) Decomposition releases carbon back to environment.

Solution: (C)

Photosynthesis does remove CO₂, nitrogen-fixing bacteria do improve soil fertility, and decomposition returns carbon to the environment—all correct. Precipitation actually reduces atmospheric water vapour, so statement C is incorrect.



Practice Questions - 4

- (i) Which sequence correctly represents movement of carbon?
 (A) Atmosphere → Plants → Animals → Atmosphere
 (B) Animals → Plants → Atmosphere
 (C) Plants → Atmosphere → Animals
 (D) Atmosphere → Animals → Plants
- (ii) Which cycle would be most directly affected by prolonged drought?
 (A) Only Carbon cycle (B) Only Nitrogen cycle
 (C) Water cycle first, then others indirectly (D) None
- (iii) Assertion (A): Nitrogen cannot be directly used by most organisms.
 Reason (R): Atmospheric nitrogen is chemically stable.
 (A) Both A and R are true, and R explains A (B) Both are true, but R does not explain A
 (C) A is true, R is false (D) A is false, R is true
- (iv) If condensation suddenly stopped occurring in nature, which immediate result would follow?
 (A) Increase in soil fertility (B) No cloud formation
 (C) Increased nitrogen fixation (D) Increase in respiration

- (v) Match the process with its function:

Process	Function
(a) Evaporation	(i) Releases carbon dioxide
(b) Respiration	(ii) Converts water into vapor
(c) Nitrogen fixation	(iii) Converts N_2 into usable compounds

Choose correct match:

- (A) a–ii, b–i, c–iii (B) a–ii, b–iii, c–i
 (C) a–i, b–iii, c–ii (D) a–i, b–ii, c–iii
- (vi) Which cycle is directly dependent on solar energy, biological activity, and atmospheric interaction simultaneously?
 (A) Only Water cycle (B) Only Nitrogen cycle
 (C) Only Carbon cycle (D) All three cycles
- (vii) A village depends on a nearby lake for fresh water. Explain how the natural water cycle prevents this lake from drying permanently.
- (viii) A scientist states that animals are temporary holders of carbon. Justify this statement using the carbon cycle.
- (ix) A region experiences heavy deforestation. Over time, carbon dioxide levels in the atmosphere increase, rainfall patterns become irregular, and soil fertility declines.
 Explain how deforestation affects the carbon cycle.
- (x) If decomposers suddenly disappeared from an ecosystem, predict the impact on both carbon and nitrogen cycles.

5. MICROORGANISMS: INVISIBLE REGULATORS OF LIFE

5.1 Beneficial Roles

Microorganisms play vital roles in maintaining ecological balance. Decomposers break down dead plants and animals into simpler substances, recycling nutrients back into the soil. Nitrogen-fixing bacteria improve soil fertility.

Certain bacteria in the human intestine help in digestion and vitamin production.

Microbes are also used in food production processes such as fermentation to prepare curd, bread, and other foods.

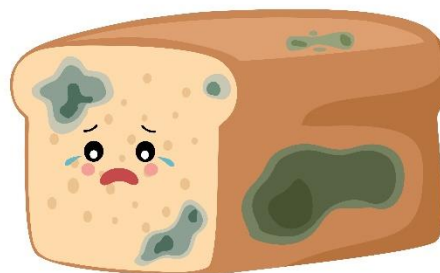


5.2 Harmful Roles

Some microorganisms are pathogenic and cause diseases in humans, animals, and plants. They can spread through air, water, food, direct contact, or insect vectors.

Preventive measures such as hygiene; safe drinking water, proper cooking, and vaccination help control infections.

Vaccines prepare the immune system to recognize and fight specific pathogens effectively.



5.3 Microbes and Environmental Balance

Microorganisms regulate nutrient availability and decomposition rates. Without decomposers, dead organic matter would accumulate and essential nutrients would not return to the soil, disrupting ecosystems.

Examples

- (i) **Compare the ecological consequences of removing decomposers and removing nitrogen-fixing bacteria.**

Solution: If decomposers were removed, dead organic matter would accumulate and nutrients would not recycle, disrupting ecosystems.

If nitrogen-fixing bacteria were removed, soil nitrogen levels would decline, reducing plant growth. Both are essential, but decomposers affect overall nutrient recycling, while nitrogen-fixing bacteria specifically affect soil fertility.

- (ii) **Suggest logical reasons why maintaining hygiene can break the chain of infection.**

Solution: Hygiene removes or reduces pathogens from hands, surfaces, and food. Clean water and proper sanitation prevents contamination.

By interrupting transmission routes such as contact and ingestion, hygiene stops the spread of infectious microorganisms.

- (iii) **Identify the incorrect statement:**

(A) Nitrogen-fixing bacteria improve soil fertility.

(B) Pathogens always benefit human health.

(C) Decomposers recycle nutrients.

(D) Vaccines stimulate immune response.

Solution: (B)

Pathogens are disease-causing microorganisms and do not benefit human health. The other statements are correct because nitrogen-fixing bacteria enrich soil, decomposers recycle nutrients, and vaccines trigger immunity.

- (iv) **Which situation best shows mutual benefit between humans and microorganisms?**

(A) Food poisoning after eating stale food

(B) Bacteria producing vitamins in the human intestine

(C) Spread of malaria by mosquitoes

(D) Crop infection by fungus

Solution: (B)

Certain bacteria in the human intestine synthesize vitamins and aid digestion, benefiting both humans and microbes. The other options describe harmful interactions, not mutual benefit.

- (v) **If decomposers disappear, which will occur first?**

(A) Increased plant growth

(B) Faster nutrient cycling

(C) Accumulation of dead matter

(D) Increased immunity

Solution: (C)

Decomposers break down dead organisms and recycle nutrients. If they disappear, dead organic matter would accumulate first because decomposition would stop.



Practice Questions - 5

- (i) Which type of microorganism decomposes dead matter?
 (A) Pathogens (B) Decomposers (bacteria & fungi)
 (C) Viruses (D) Protozoa
- (ii) Which factor most directly controls the spread of microbial diseases in a city?
 (A) Soil fertility (B) Vaccination and hygiene
 (C) Number of decomposers (D) Fermentation process
- (iii) Which of the following statements is correct?
 (A) All bacteria are harmful.
 (B) Microorganisms have no role in agriculture.
 (C) Some microbes protect crops from diseases.
 (D) Vaccines cure existing infections immediately.
- (iv) Identify the odd one out:
 (A) *Lactobacillus* (B) Yeast
 (C) Virus (D) Nitrogen-fixing bacteria
- (v) Assertion (A): Vaccination reduces the spread of infectious diseases.
 Reason (R): Vaccines destroy all microbes present in the body.
 (A) Both A and R are true (B) Both A and R are false
 (C) A is true, R is false (D) A is false, R is true
- (vi) Which activity best represents nutrient recycling?
 (A) Vaccination (B) Fermentation
 (C) Decomposition (D) Respiration
- (vii) How do microorganisms help in maintaining soil structure?
- (viii) A community stops following vaccination programs. Predict three possible outcomes.
- (ix) "Microorganisms are both friends and enemies of humans." Explain this statement with suitable reasoning.
- (x) Justify the statement: "Microorganisms are essential for nutrient recycling in nature."

6. HUMAN HEALTH AS BIOLOGICAL STABILITY

6.1 Nutrition and Immunity

A balanced diet provides carbohydrates, proteins, fats, vitamins, and minerals necessary for growth and repair. Micronutrients such as vitamins and minerals support immune function. Adequate nutrition strengthens the body's ability to produce antibodies and resist infections.

6.2 Circulation and Transport

The circulatory system transports oxygen from the lungs to body cells and carries nutrients from the digestive system. It also removes carbon dioxide and other waste products. Efficient circulation ensures proper functioning of all organs.

6.3 Excretion and Detoxification

Metabolic processes produce waste substances such as urea and carbon dioxide. The excretory system removes these wastes to prevent accumulation and toxicity. Kidneys filter blood and maintain water and salt balance. Excretion helps maintain internal chemical stability.

6.4 Hygiene and Disease Prevention

Clean surroundings reduce microbial growth and disease transmission.

Proper sanitation, safe drinking water, and personal hygiene prevent water-borne and communicable diseases. Environmental cleanliness directly influences public health.



Examples

- (i) **Rahul, a 12-year-old, often falls sick and feels weak. His diet mainly consists of fried snacks and sugary drinks. He rarely eats fruits, vegetables, or protein-rich foods.**

Identify two reasons why Rahul is more prone to infections.

Solution: **Poor diet lacking essential nutrients:** Rahul's diet mainly has fried snacks and sugary drinks, with very few fruits, vegetables, or protein-rich foods. This means he is not getting vitamins, minerals, and proteins needed to strengthen his immune system, making him more susceptible to infections.

Weakened body resistance: The lack of a balanced diet reduces the body's ability to produce antibodies, which are crucial to fight microbes. As a result, his body cannot resist infections effectively.

- (ii) **Discuss the importance of hygiene and clean surroundings in preventing diseases.**

Solution: Hygiene and clean surroundings play a major role in preventing infections. Microbes, which cause communicable and water-borne diseases, grow easily in unclean environments. By maintaining personal hygiene (like washing hands regularly) and keeping the surroundings clean (like proper sanitation and safe drinking water), the growth and spread of harmful microbes are reduced. This directly prevents the transmission of diseases, protects health, and promotes overall public health.

- (iii) **Which statement correctly describes the role of a balanced diet?**

(A) It provides only energy for daily activities.

(B) It includes carbohydrates, proteins, fats, vitamins, and minerals for growth and repair.

(C) It consists only of vitamins and minerals to strengthen immunity.

(D) It focuses mainly on water and electrolytes.

Solution: (B)

A balanced diet supplies carbohydrates, proteins, fats, vitamins, and minerals needed for energy, growth, repair, and proper body functioning. It is not limited to only one type of nutrient.

- (iv) **Maintaining hygiene and clean surroundings prevents:**

(A) Digestive enzyme deficiency

(B) Microbial growth and disease transmission

(C) Fat accumulation in cells

(D) Vitamin deficiency

Solution: (B)

Good hygiene and clean surroundings reduce the growth and spread of harmful microorganisms. This helps prevent infections and disease transmission.

- (v) **Assertion (A): Kidneys are essential for maintaining chemical balance in the body.**

Reason (R): Kidneys excrete urea and regulate water and salt levels.

(A) Both A and R are true, and R explains A.

(B) Both A and R are true, but R does not explain A.

(C) A is true, R is false.

(D) A is false, R is true.

Solution: (A)

Kidneys maintain the body's chemical balance by removing urea and regulating water and salt levels in the blood. Thus, both statements are true and the reason correctly explains the assertion.



Practice Questions - 6

- (i) Identify the correct statements regarding the circulatory system.
1. It transports oxygen from lungs to body cells.
 2. It carries nutrients from the digestive system to cells.
 3. It eliminates urea directly from blood.
 4. Efficient circulation ensures proper functioning of all organs.
- (A) 1, 2, 4 (B) 2, 3, 4
(C) 1, 3, 4 (D) All 1, 2, 3, 4
- (ii) Assertion (A): Environmental cleanliness reduces the spread of infectious diseases.
Reason (R): Microbes cannot survive in unclean surroundings.
- (A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, R is false.
(D) A is false, R is true.
- (iii) Which does NOT directly prevent water-borne diseases?
- (A) Safe drinking water (B) Clean surroundings
(C) Handwashing (D) Balanced diet
- (iv) Proper hygiene reduces:
- (A) Microbial growth (B) Communicable diseases
(C) Nutrient absorption (D) Promotes public health
- (v) A person shows swelling in feet and reduced urine output. Likely problem?
- (A) Circulatory system (B) Excretory system
(C) Digestive system (D) Immune system
- (vi) Assertion (A): Proper nutrition increases the body's ability to fight infections.
Reason (R): Proteins help in the formation of antibodies.
- (A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, R is false.
(D) Both A and R are false.
- (vii) How does a balanced diet help the body resist infections?
- (viii) A school child frequently falls ill. Suggest two nutrition-related measures to improve immunity and explain why.
- (ix) Describe the importance of excretion in keeping the body's internal environment stable.
- (x) Explain how the circulatory system works to maintain healthy body function.

7. EARTH AS A LIFE-SUPPORTING PLANET AND ECOSYSTEM INTERDEPENDENCE

7.1 Biotic and Abiotic Components

Ecosystems are composed of both living (biotic) and non-living (abiotic) components, which continuously interact to maintain balance and stability. The biotic components include plants, animals, and microorganisms, each performing essential roles. Plants produce food through photosynthesis, herbivores feed on plants, and carnivores feed on herbivores, forming the foundation of energy flow in ecosystems.

Microorganisms, including bacteria and fungi, are crucial for decomposition, breaking down dead plants and animals into nutrients that enrich the soil, ensuring that plants can grow and the nutrient cycle continues. The abiotic components, such as air, water, soil, sunlight, and temperature, provide the environmental conditions necessary for survival.

Air supplies oxygen for respiration and carbon dioxide for photosynthesis; water supports cellular functions and transportation of nutrients; sunlight provides energy for producers; and soil provides minerals and a habitat for many organisms. The interaction between biotic and abiotic factors ensures that ecosystems remain stable, self-sustaining, and resilient to minor changes, forming a dynamic balance that supports life.

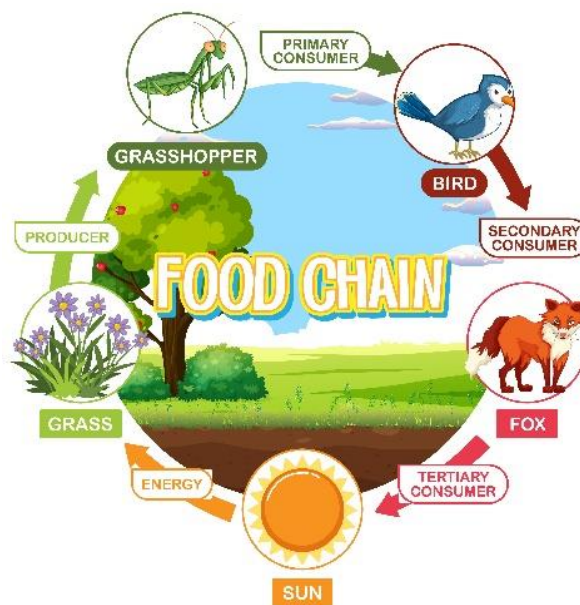
7.2 Food Chains and Food Webs

Energy flows in ecosystems in a structured manner, beginning with producers and moving through consumers, forming food chains. A simple food chain might involve grass → rabbit → fox, where each organism depends on the previous one for energy. However, ecosystems are rarely simple.

Most organisms have multiple feeding relationships, which form food webs, where several interconnected food chains create a complex network of energy flow. Food webs ensure that if one species declines due to disease, habitat loss, or predation, others can compensate, allowing energy to continue flowing through the ecosystem.

For example, if rabbits decrease in number, foxes may feed on alternative herbivores like mice, and grass may grow more abundantly to support the next generation of herbivores.

Food webs also show how energy is lost at each trophic level through processes like respiration and heat production, emphasizing the importance of producers and decomposers in maintaining ecosystem energy balance.



7.3 Trophic Levels and Interdependence

Organisms occupy specific trophic levels, which define their position in the ecosystem's feeding hierarchy. Producers such as grasses, trees, and algae form the base by capturing solar energy through photosynthesis, producing food that sustains all other organisms.

Primary consumers, or herbivores, feed directly on producers, while secondary and tertiary consumers, or carnivores, feed on herbivores or other carnivores. Decomposers like fungi and bacteria occupy an essential role, breaking down dead matter and returning nutrients to the soil.

The ecosystem demonstrates interdependence, as the survival of each organism depends on others. For example, a decline in herbivores affects carnivores by reducing their food supply and affects producers by allowing overgrowth. This interconnectedness highlights the delicate balance of ecosystems and the importance of maintaining biodiversity, as the loss of one species can have cascading effects on the entire system.

7.4 Atmospheric Gases and Life Support

The atmosphere is essential for sustaining life by supplying gases and creating conditions suitable for survival. Oxygen is necessary for the respiration of animals and humans, enabling energy production from food. Carbon dioxide is required by plants for photosynthesis, allowing the production of carbohydrates, which serve as energy for herbivores and carnivores.

Nitrogen, though inert in the atmosphere, becomes usable for plants through processes such as nitrogen fixation by bacteria, supporting plant growth and nutrient cycling. Beyond gases, the atmosphere regulates temperature, protecting organisms from extreme heat and cold, and acts as a shield against harmful solar radiation.

It supports weather and climate patterns, which influence the distribution of species, water availability, and seasonal cycles in ecosystems. Without the atmosphere's protective and regulatory roles, life as we know it could not exist.

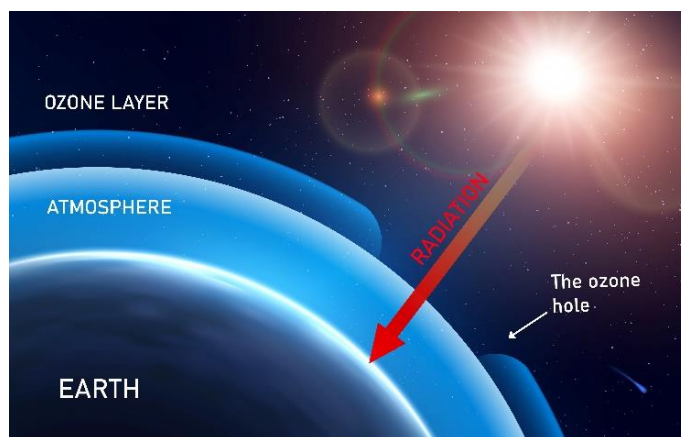
7.5 Ozone Layer Protection

The ozone layer is a critical component of the atmosphere, located in the stratosphere. It absorbs harmful ultraviolet (UV) radiation from the Sun, preventing it from reaching Earth's surface in dangerous amounts.

Excessive UV radiation can damage the DNA and cells of living organisms, causing mutations, cancer, and reduced reproductive success.

Plants, which form the basis of food chains, can also be harmed, potentially disrupting entire ecosystems. The ozone layer thus maintains ecosystem stability by shielding life from radiation-related damage.

Human activities, such as the use of chlorofluorocarbons (CFCs) and certain industrial chemicals, have depleted the ozone layer, highlighting the importance of conservation measures to protect this natural shield and ensure continued survival of terrestrial life.



7.6 Temperature Regulation

Earth's moderate distance from the Sun ensures that temperatures remain suitable for the presence of liquid water, a necessity for all known forms of life. Temperature regulation affects metabolic rates, growth, reproduction, and survival of organisms.

Extreme temperatures can disrupt water availability, cause heat stress or freezing, and even lead to the collapse of local ecosystems. Stable temperatures, combined with other factors like nutrient availability, oxygen, and sunlight, allow complex life forms to thrive and ecosystems to remain productive and balanced.

Temperature interacts with other abiotic and biotic factors, influencing plant growth, animal behavior, food web dynamics, and species distribution, emphasizing its central role in sustaining life on Earth.

Examples

- (i) **A farmer notices that after eliminating snakes from his farm, the number of rodents increased dramatically, causing crop loss. Explain this using food web concepts.**

Solution: Snakes are carnivores that control rodent populations. Removing snakes breaks the food web, allowing rodent numbers to increase unchecked. Crops, which are the primary producers, get consumed excessively, showing how changes at one level affect other organisms and the stability of the ecosystem.

- (ii) **Discuss the role of microorganisms in maintaining ecosystem stability and what would happen if they were removed.**

Solution: Microorganisms decompose dead plants and animals, recycling nutrients into the soil. Without them, nutrients would not return to the soil efficiently, causing soil infertility, reduced plant growth, and less food for herbivores and carnivores. This would disrupt energy flow, reduce biodiversity, and destabilize the ecosystem, showing microorganisms' essential role in maintaining balance.

- (iii) **Which abiotic factor directly affects temperature regulation in ecosystems?**

(A) Soil

(B) Sunlight

(C) Herbivores

(D) Microorganisms

Solution: (B)

Sunlight is the primary abiotic factor controlling temperature in ecosystems because it provides heat energy to Earth's surface. Soil, herbivores, and microorganisms do not directly regulate temperature.

- (iv) **Select the true statements about atmospheric gases:**

1. Oxygen supports respiration

2. Carbon dioxide supports photosynthesis

3. Nitrogen is fixed by certain bacteria for plant use

4. Methane regulates ecosystem energy flow

(A) 1, 2, 3

(B) 1, 3, 4

(C) 2, 3, 4

(D) 1, 2, 4

Solution: (A)

Oxygen supports respiration, carbon dioxide is used in photosynthesis, and certain bacteria fix atmospheric nitrogen for plant use—all true. Methane does not regulate ecosystem energy flow, so statement 4 is false.

- (v) **Assertion (A): Oxygen is necessary for the respiration of animals and humans.**

Reason (R): Carbon dioxide is required by plants for respiration.

(A) Both A and R are true, and R explains A.

(B) Both A and R are true, but R does not explain A.

(C) A is true, R is false.

(D) A is false, R is true.

Solution: (C)

Respiration uses O_2 and releases CO_2 (in both plants and animals); photosynthesis uses CO_2 and releases O_2 (in plants).



PRACTICE QUESTIONS

- (i) Earth's moderate distance from the Sun is essential because it:
 - (A) Provides sunlight for decomposers only.
 - (B) Maintains stable temperatures for water and life.
 - (C) Increases carnivore populations.
 - (D) Eliminates herbivore overpopulation.

- (ii) If herbivore populations increase excessively, which of the following is most likely?
 - (A) Abiotic components are destroyed.
 - (B) Carnivore population decreases.
 - (C) Decomposer activity stops.
 - (D) Producer population decreases.

- (iii) Assertion (A): Ozone layer depletion indirectly affects food chains.
 Reason (R): Increased UV radiation can reduce plant productivity, impacting herbivores and carnivores.
 - (A) Both A and R are true, and R is the correct explanation of A.
 - (B) Both A and R are true, but R is not the correct explanation of A.
 - (C) A is true, R is false.
 - (D) A is false, R is true.

- (iv) Which of the following is true about abiotic components?
 - (A) They produce food for herbivores
 - (B) They provide sunlight, water, and minerals necessary for life
 - (C) They consume dead organisms
 - (D) They form trophic levels

- (v) In the food chain: Grass → Grasshopper → Frog → Snake, which is the secondary consumer?
 - (A) Grass
 - (B) Grasshopper
 - (C) Frog
 - (D) Snake

- (vi) Choose correct statements about trophic levels:
 - 1. Producers capture solar energy
 - 2. Primary consumers feed on producers
 - 3. Secondary consumers feed on other carnivores only
 - 4. Decomposers recycle nutrients to producers
 - (A) 1, 2, 4
 - (B) 1, 3
 - (C) 2, 3, 4
 - (D) 1, 2, 3

- (vii) Compare the roles of producers and decomposers in ecosystem stability. Which one is more critical if a food web loses a primary consumer, and why?

- (viii) Discuss the importance of the ozone layer in protecting ecosystems and the effects of human activities.

- (ix) How does Earth's moderate distance from the Sun contribute to the stability and productivity of ecosystems?

- (x) Explain how Earth's moderate distance from the Sun contributes to temperature regulation and ecosystem stability. What might happen if temperatures were significantly higher or lower?

Practice Questions - Answer Key

- (1) (i) (C), (ii) (B), (iii) (C), (iv) (A), (v) (B), (vi) (B).**
- (2) (i) (C), (ii) (C), (iii) (C), (iv) (C), (v) (A), (vi) (B).**
- (3) (i) (B), (ii) (A), (iii) (C), (iv) (C), (v) (C), (vi) (C).**
- (4) (i) (A), (ii) (C), (iii) (A), (iv) (B), (v) (A), (vi) (D).**
- (5) (i) (B), (ii) (B), (iii) (C), (iv) (C), (v) (C), (vi) (C).**
- (6) (i) (A), (ii) (C), (iii) (D), (iv) (B), (v) (B), (vi) (A).**
- (7) (i) (B), (ii) (D), (iii) (A), (iv) (B), (v) (C), (vi) (A).**